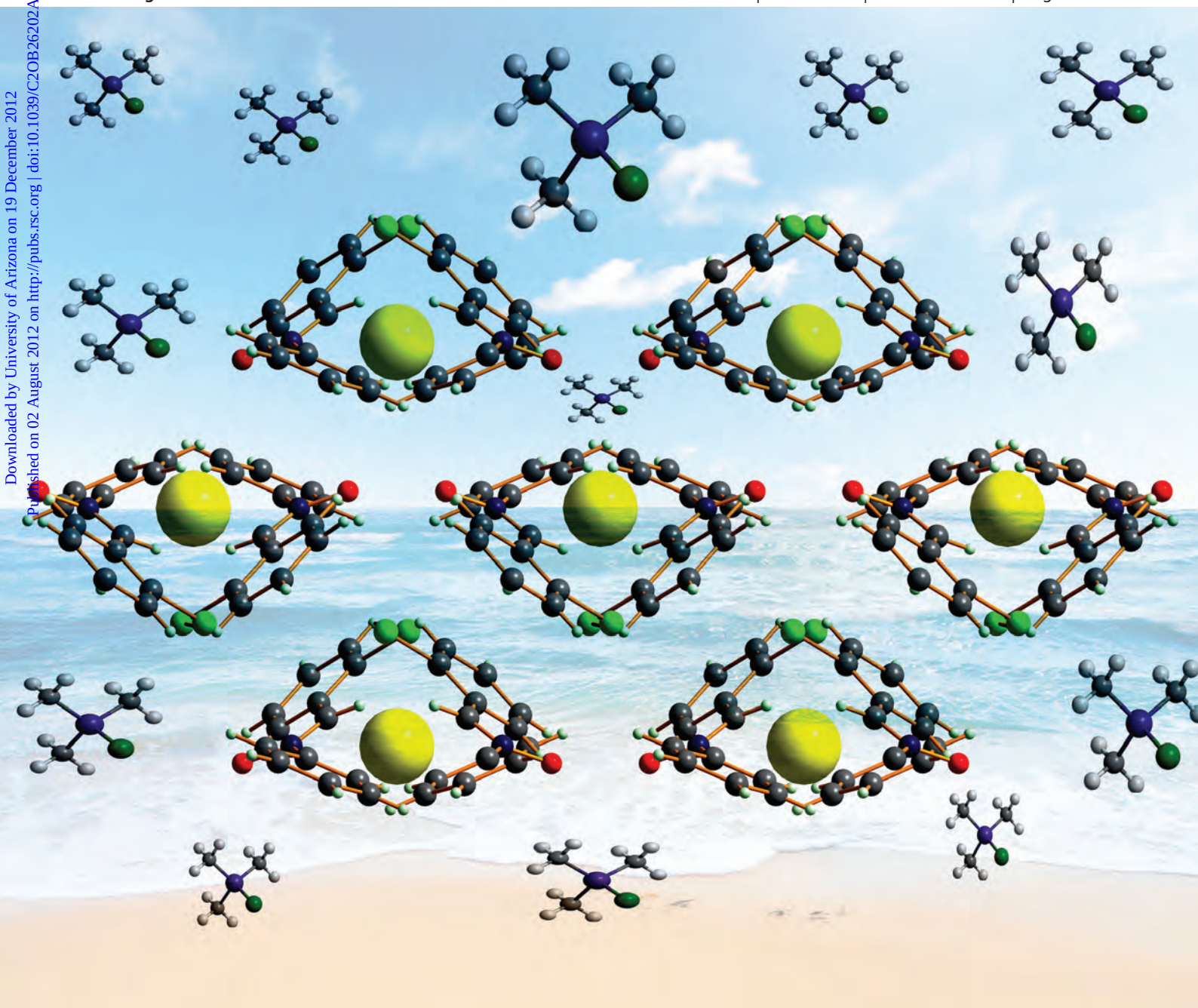


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COMMUNICATION

Trimethylsilyl chloride promoted synthesis of α -branched amines by nucleophilic addition of organozinc halides to nitrones†Ying Fu,*^a Yanhua Liu,^a Yaojuan Chen,^a Helmut M. Hügel,^b Minzhu Wang,^a Danfeng Huang^a and Yulai Hu^a

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A general procedure for the nucleophilic addition of organozinc halides with nitrones in the presence of trimethylsilyl chloride has been developed. Trimethylsilyl chloride was found to be both an indispensable reaction promoter and a ready hydroxylamine protective agent in these reactions. The produced *O*-(trimethylsilyl)hydroxylamines can be easily reduced into corresponding amines just by a zinc–copper couple in saturated aqueous NH_4Cl solution.

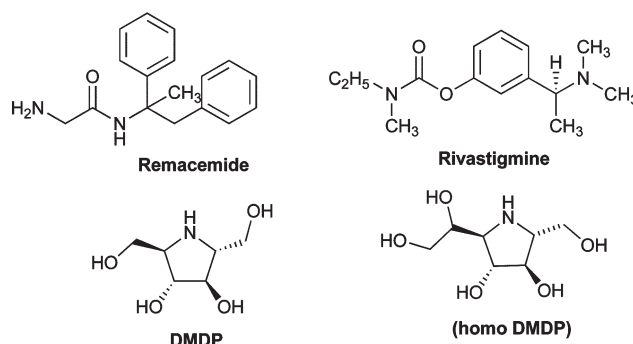
α -Branched amines represent an important class of nitrogen containing compounds, due to their attractive biological and pharmacological activities. For example, Remacemide is used for the treatment of acute ischemic stroke, epilepsy, Huntington's disease and Parkinson's disease.¹ Rivastigmine is a reversible cholinesterase inhibitor and is used to treat Alzheimer's disease.² In addition, α -branched amines are highly valuable molecules in bioorganic chemistry as they are widely found in nature, such as 2,5-dideoxy-2,5-imino-D-mannitol (DMDP) and 2,5-dideoxy-2,5-imino-DL-glycero-D-manno-heptitol (homo DMDP), which are found to be efficient glycoside inhibitors.³ Thus, efficient syntheses of a library of α -branched amines are of interest in both organic and medicinal chemistry (Fig. 1).⁴

The nucleophilic addition reaction of organometallic reagents to nitrones is a convenient and reliable procedure, and hence, is widely used in the synthesis of structurally variable α -branched amines.⁵ Grignard reagents and organolithium reagents are frequently deployed in these conversions due to their ready availability and high reactivity, especially when chiral nitrones⁶ or chiral catalysts⁷ were introduced, high yields as well as high stereoselectivities are always obtained. A large variety of natural and semi-natural products such as pyrrolizidines and indolizidines were successfully prepared *via* this protocol.⁸ However, organozinc reagents, a kind of less reactive yet more selective organometallic reagents,⁹ although extensively used in the reactions with carbonyl compounds, are less explored in reactions

with imine derivatives such as nitrones. A few reports have appeared in which some reactive organozinc species, such as diorganozinc¹⁰ or diorganozinc mediated alkynylation,¹¹ allylation,¹² vinylation¹³ and the Reformatsky reaction¹⁴ are effective. However, to the best of our knowledge, a practical synthesis of α -branched amine derivatives by nucleophilic addition of organozinc halides to nitrones has not been fully explored to date.

Organozinc halides are one of the most useful organometallic reagents that can be easily prepared by direct insertion of zinc metal into corresponding functionalized organic halides and are therefore considered as a “real kind” of functionalized organometallics.¹⁵ We were attracted by their wide application in synthetic chemistry, and have reported that reaction of benzylic zinc halides with aryl aldehydes under transition-metal-complex catalysis in the presence of trimethylsilyl chloride (TMSCl) leads to *trans*-stilbenes in high yield,¹⁶ whereas reaction of primary alkyl zinc halides with aromatic aldehydes gave α,α -dialkyltoluenes.¹⁷ To broaden the utility of organozinc reagents in organic synthesis, especially with the aim of exploring the utilization of organozinc halides as a mild and selectively functionalized nucleophile for the synthesis of α -branched amines, we report here an interesting TMSCl promoted nucleophilic addition reaction of organozinc halides with nitrones in which TMSCl was found to be both an indispensable reaction promoter and a ready hydroxylamine protective agent.

Initially, the reaction of benzylzinc chloride **1a** and *N*-(4-methoxybenzylidene)aniline oxide **2a** was examined without any catalyst. Unfortunately, no reaction occurred at room

Fig. 1 Drugs and natural products of α -branched amines.

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Table 1 Optimization of the reaction conditions^a

Entry	Temp [°C]	Time [h]	5 (equiv) ^b	3a/4a	Yield ^d [%]
1	RT	12	—	—	0
2	60	6	—	1/0	65
3	RT	6	TMSCl (1.0)	1/3 ^c	78
4	RT	0.5	TMSCl (2.0)	0/1	92
5	RT	0.5	TMSBr (2.0)	0/1	88
6	RT	0.5	TMSI (2.0)	0/1	86
7	RT	0.5	TMSOTf (2.0)	0/1	91
8	RT	6	TBDMSCl (2.0)	—	—
9	RT	6	TBDPMSCl (2.0)	—	—

^a Reaction conditions: nitron **2a** (3.0 mmol) was treated with benzylzinc chloride **1a** (4.5 mmol) and organosilyl halides in THF (15 mL) at room temperature under argon. ^b Amount of organosilyl halides was based on nitron **2a**. ^c The ratio was determined by ¹H NMR spectra. ^d Isolated yields.

temperature. When the temperature was raised to 60 °C, the reaction proceeded sluggishly and after 6 hours, the nitron **2a** had completely consumed and the corresponding hydroxylamine **3a** was obtained in 65% yield (Table 1, entries 1 and 2).

Lewis acids can greatly enhance the reactivity of nitrones in various kinds of reactions such as the 1,3-dipolar cycloaddition¹⁸ and nucleophilic addition reactions.^{8b} When one equivalent of TMSCl was added, the reaction proceeded slowly at room temperature and nitron **2a** disappeared in six hours. However, hydroxylamine **3a** and the *O*-TMS hydroxylamine ether **4a** were formed in 78% overall yield (**3a/4a** 1 : 3). Gratifying, when 2 equivalents of TMSCl were added, the reaction proceeded quite quickly and cleanly. Nitron **2a** was totally consumed in only 30 minutes and the *O*-TMS hydroxylamine ether **4a** was obtained as the only product in 92% isolated yield (Table 1, entries 3 and 4).

With the preliminary success of the effect of trimethylsilyl chloride, several other organosilyl reagents were then screened. TMSBr, TMSI and TMSOTf worked in the same way in this reaction system. They all gave the desired silylated hydroxylamine derivatives **4a** cleanly with similar yield compared to TMSCl (entries 5–7). However, the widely used hydroxyl group protection reagents, *tert*-butyldimethylsilyl chloride (TBDMSCl) and *tert*-butyldiphenylsilyl chloride (TBDPMSCl) were ineffective (entries 8 and 9). When considering the price and availability of these trimethylsilyl reagents, TMSCl is the best choice and was selected in our following studies.

Attracted by this unexpected reactivity, we then investigated the efficacy of TMSCl on reactions of various benzylic zinc halides with α,N -diaryl nitrones. Different kinds of α,N -diaryl nitrones bearing both electron donating and electron withdrawing groups were subjected to the optimized reaction conditions (Table 2). The results showed that the reaction has broad applicability. *N*-Phenyl nitrones bearing both electron-donating and electron-withdrawing groups on the *C*-phenyl rings reacted with

Table 2 Reaction of benzylic zinc chlorides with α,N -diaryl nitrones^a

Entry	Ar	R	Yield ^b (%)
1	Ph (1a)	4-CH ₃ OC ₆ H ₄ (2a)	92 (4a)
2	Ph (1a)	Ph (2b)	90 (4b)
3	Ph (1a)	4-CH ₃ C ₆ H ₄ (2c)	86 (4c)
4	Ph (1a)	3,4,5-(CH ₃ O) ₃ C ₆ H ₂ (2d)	75 (4d)
5	Ph (1a)	4-HOC ₆ H ₄ (2e)	75 ^c (4e)
6	Ph (1a)	3-HO-4-CH ₃ OC ₆ H ₃ (2f)	67 ^c (4f)
7	Ph (1a)	2-ClC ₆ H ₄ (2g)	70 (4g)
8	Ph (1a)	4-ClC ₆ H ₄ (2h)	82 (4h)
9	Ph (1a)	2,4-Cl ₂ C ₆ H ₃ (2i)	91 (4i)
10	Ph (1a)	2-Thienyl (2j)	80 (4j)
11	Ph (1a)	2-Furanyl (2k)	74 (4k)
12	4-ClC ₆ H ₄ (1b)	Ph (2b)	86 (4l)
13	4-ClC ₆ H ₄ (1b)	4-CH ₃ OC ₆ H ₄ (2a)	88 (4m)
14	4-ClC ₆ H ₄ (1b)	3,4,5-(CH ₃ O) ₃ C ₆ H ₂ (2d)	83 (4n)
15	4-ClC ₆ H ₄ (1b)	2,4-Cl ₂ C ₆ H ₃ (2i)	82 (4o)
16	4-ClC ₆ H ₄ (1b)	2-Thienyl (2j)	79 (4p)
17	2-ClC ₆ H ₄ (1c)	Ph (2b)	73 (4q)
18	2-Naphthyl (1d)	4-ClC ₆ H ₄ (2h)	62 (4r)

^a Unless otherwise noted, the reaction were performed by employing benzylic zinc chloride (4.5 mmol), *N*, α -diphenyl nitron (3 mmol) in THF (15 mL). ^b Isolated yield. ^c Benzylic zinc halide (2.5 equiv).

benzylic zinc halide to give the corresponding *O*-silylated hydroxylamines in moderate to high yields. Aromatic heterocyclic aldehyde derived nitrones such as 2-thienyl and 2-furanyl substituted nitrones all reacted smoothly, with the corresponding silylated hydroxylamine derivatives obtained in good yield (Table 2, entries 10, 11 and 16). Interestingly, hydroxyl substituents on the benzyldine ring of the nitrones do not require further protection as the desired *O*-silylated hydroxylamine ethers were obtained in moderate yield when a 2.5 molar ratio of benzylic zinc chloride was added (Table 2, entries 5 and 6).

Encouraged by our success with benzylic zinc chlorides, we then investigated other organozinc reagents (Table 3). To our delight, other organozinc reagents (R¹ZnBr) selected, such as R¹ = aryl (entries 1–6), thienyl (entries 7–9) and alkyl zinc bromides (entries 10–15) all reacted smoothly with α,N -diaryl nitrones and gave the desired products in good yields. Both primary and secondary alkyl zinc bromide can be applied here (entries 10–15). Halogens such as chloro, fluoro are tolerated in both organozinc halides and nitrones. However, nitrones bearing hydroxyl groups were unreactive to aryl and alkyl zinc reagents this time (entries 16 and 17). This is probably due to the strong electron donating effects of the phenolate of the hydroxylated nitron¹⁹ which significantly reduced the electrophilicity of these nitrones so that only the more reactive organozinc species such as benzylic zinc halide could be used successfully.

Although Grignard reagents can react readily with nitrones and give hydroxylamines in high yield,²⁰ our TMSCl promoted organozinc reagent method has some significant advantages. The organozinc reagent can be highly functionalized whereas functionalized Grignard reagents are not readily utilized. The most practical advantage of our protocol is that the reaction of

Table 3 Reaction of various organozinc bromides with nitrones^a

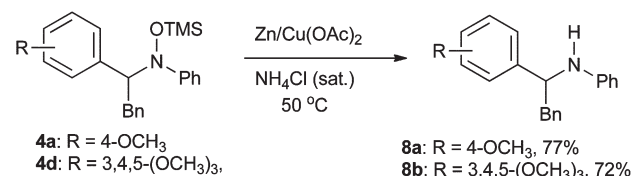
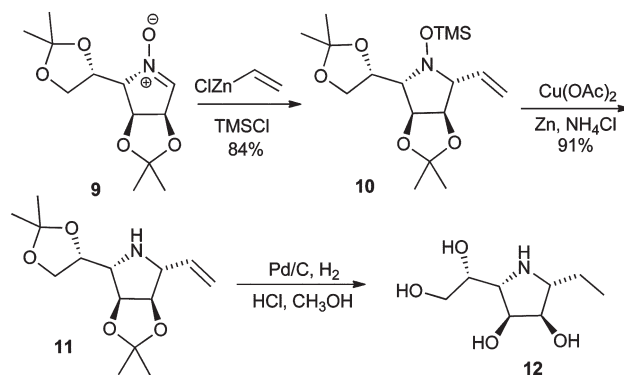
Entry	7 R ¹	2 R ²	R ³	Yield ^b (%)
1	Ph(7a)	Ph(2b)	Bn	73 (6a)
2	Ph(7a)	3,4,5-(CH ₃ O) ₃ C ₆ H ₂ (2d)	Ph	91 (6b)
3	Ph(7a)	2-ClC ₆ H ₄ (2g)	Ph	71 (6c)
4	Ph(7a)	4-FC ₆ H ₄ (2l)	Ph	84 (6d)
5	2-ClC ₆ H ₄ (7b)	Ph(2b)	Ph	86 (6c)
6	1-Naphthyl(7c)	4-CH ₃ C ₆ H ₄ (2c)	Ph	73 (6e)
7	3-Thienyl(7d)	2-ClC ₆ H ₄ (2g)	Ph	76 (6f)
8	3-Thienyl(7d)	4-FC ₆ H ₄ (2l)	Ph	77 (6g)
9	3-Thienyl(7d)	2,4-Cl ₂ C ₆ H ₃ (2i)	Ph	82 (6h)
10	<i>n</i> -Hexyl(7e)	4-CH ₃ OC ₆ H ₄ (2a)	Ph	77 (6i)
11	<i>n</i> -Hexyl(7e)	3,4,5-(CH ₃ O) ₃ C ₆ H ₂ (2d)	Ph	75 (6j)
12	<i>n</i> -Hexyl(7e)	2-ClC ₆ H ₄ (2g)	Ph	79 (6k)
13	<i>n</i> -Hexyl(7e)	2-Thienyl(2j)	Ph	75 (6l)
14	<i>s</i> -Butyl(7f)	4-CH ₃ C ₆ H ₄ (2c)	Ph	65 (6m)
15	<i>s</i> -Butyl(7f)	2-CH ₃ OC ₆ H ₄ (2n)	Ph	75 (6n)
16	ph(7a)	3-HO-4-CH ₃ OC ₆ H ₃ (2f)	Ph	NR ^c
17	<i>n</i> -Hexyl(7e)	4-HOC ₆ H ₄ (2e)	Ph	NR ^c

^a Unless otherwise noted, the reaction was performed by employing organozinc halide (4.5 mmol), *N*, α -diphenyl nitrone (3 mmol) in THF (15 mL). ^b Isolated yield. ^c NR = No reaction.

Grignard reagents with nitrones leads only to hydroxylamines under the same reaction condition and since the hydroxylamine derivatives, especially *N*-substituted aromatic hydroxylamines are very air sensitive. They decompose readily²⁰ and usually cannot be isolated in a pure form by quickly polymerizing after column chromatography, thus, the synthesis, purification and characterization of hydroxylamines proved challenging. Even when these compounds were kept under nitrogen in a deep freezer for several weeks, they were found to have decomposed as indicated by ¹H NMR analysis. However, in our protocol, the *O*-TMS hydroxylamine ethers are stable to air and can be kept in a refrigerator for months without any decomposition. We consider our protocol whereby TMSCl serves as a reaction promoter and a ready *N*-hydroxylamine protecting reagent is a significant improvement on the reaction of any other previously reported Grignard and diorganozinc methods.

The *O*-trimethylsilyl hydroxylamine ethers can be easily converted into corresponding amines by simply stirring the ethers with a zinc–copper couple in a saturated aqueous NH₄Cl at 50 °C for 1 hour. As illustrated in Scheme 1, the substrates **4a** and **4d** were reduced using this protocol to the corresponding amines in 77% and 72% isolated yields respectively.

To test the application of our TMSCl promoted organozinc protocol for natural product synthesis, we prepared a cyclic nitron **9** from D-mannose²¹ (Scheme 2). Gratifyingly, **9** reacted with vinylzinc chloride in the presence of two equivalents of TMSCl at room temperature to furnish the desired *O*-TMS-hydroxylamine ether **10** in 84% yield stereospecifically. Upon further stirring the reaction mixture at room temperature for

**Scheme 1** Reduction of *O*-TMS hydroxylamine ethers to amines.**Scheme 2** Synthesis of α -branched polyhydroxylated pyrrolidine **12**.

2 hours in saturated aqueous NH₄Cl solution and with zinc metal in the presence of catalytic amount of Cu(OAc)₂, **10** was completely reduced to the pyrrolidine **11** in 91% yield. Deprotection of **11** by catalytic hydrogenation in acid solution gave the α -ethyl polyhydroxylated pyrrolidine **12** in quantitative yield which is a structural analogue of 2,5-dideoxy-2,5-imino-DL-glycero-D-manno-heptitol (homo DMDP), a natural selective inhibitor of β -glucosidases.^{3b}

In conclusion, we have successfully developed a new method for the synthesis of substituted *O*-trimethylsilyl hydroxylamine ethers via a TMSCl-catalyzed nucleophilic addition of organozinc halides to nitrones. To the best of our knowledge, the present reaction is the first example of a TMSCl catalyzed nucleophilic addition reaction of organozinc reagents with nitrones whereby the TMSCl acts both as a reaction promoter and a good protection reagent for the sensitive *N*-hydroxylamine group. Furthermore, the *O*-TMS protected hydroxylamine ethers can be easily reduced by zinc to the corresponding amines. This protocol offers a practically useful method in natural product synthesis.

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